

Week 3 — Task 9: Network Traffic Analysis — Wireshark Basics

1. Introduction

Network traffic analysis is a core SOC analyst skill. It involves examining packets and network protocols to understand communication between systems and identify potentially suspicious activity.

This project focused on Wireshark basics, packet captures, display filters, DNS, HTTP, TCP, IP addresses, and basic security analysis.

2. Objectives

The objectives were to:

- Understand packet captures.
- Identify common network protocols.
- Use Wireshark display filters.
- Filter traffic by IP address.
- Identify DNS traffic.
- Identify HTTP traffic.
- Identify TCP traffic.
- Inspect an HTTP request.
- Understand plaintext information exposure.
- Recognize suspicious network indicators.

3. Packet Capture

The packet capture used during the practical analysis was:

MCPE-0.15.pcapng

The capture contained approximately 120 packets.

Packet captures provide a record of network communication that can be examined during security investigations.

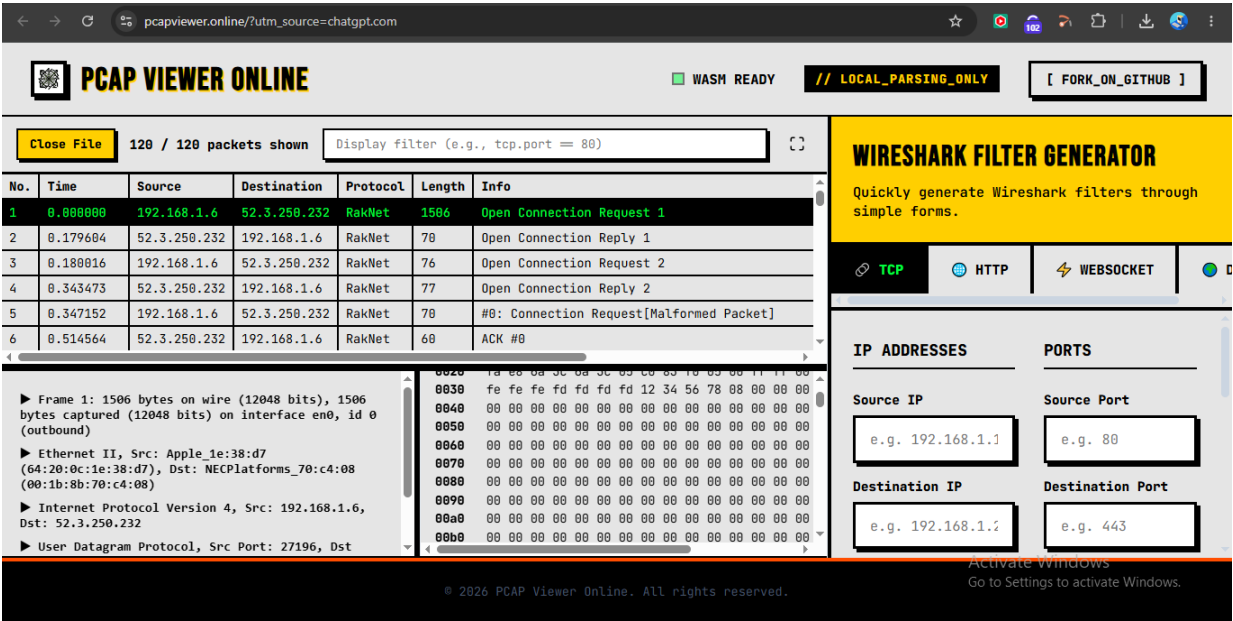
An analyst can inspect information such as:

- Source IP
- Destination IP

- Source port
- Destination port
- Protocol
- Packet length
- TCP flags
- DNS queries
- HTTP requests
- HTTP responses

4. Screenshot Evidence

Screenshot 1



Screenshot 2

PCAP VIEWER ONLINE

WASH READY

// LOCAL_PARSING_ONLY

[FORK_ON_GITHUB]

Close File

120 / 120 packets shown

ip.addr == 192.168.1.6

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.1.6	52.3.250.232	RakNet	1506	Open Connection Request 1
2	0.179604	52.3.250.232	192.168.1.6	RakNet	70	Open Connection Reply 1
3	0.180016	192.168.1.6	52.3.250.232	RakNet	76	Open Connection Request 2
4	0.343473	52.3.250.232	192.168.1.6	RakNet	77	Open Connection Reply 2
5	0.347152	192.168.1.6	52.3.250.232	RakNet	78	#0: Connection Request[Malformed Packet]
6	0.514564	52.3.250.232	192.168.1.6	RakNet	60	
7	0.526991	52.3.250.232	192.168.1.6	RakNet	152	#0: Connection Request Accepted
8	0.527251	192.168.1.6	52.3.250.232	RakNet	49	
9	0.538728	192.168.1.6	52.3.250.232	RakNet	162	#1: New Incoming Connection, Connected Ping

▶ Frame 1: 1506 bytes on wire (12048 bits), 1506 bytes captured (12048 bits) on interface en0, id 0 (outbound)

▶ Ethernet II, Src: Apple_1e:38:d7 (64:2b:8c:1e:38:d7), Dst: NECPlatforms_70:c4:08

0020 1a ee 0a 3c 0a 3c 05 c0 05 10 05 00 11 11 00

0030 fe fe fe fd fd fd fd 12 34 56 78 00 00 00 00

0040 00 00 00 00 00 00 00 00 00 00 00 00 00 00

0050 00 00 00 00 00 00 00 00 00 00 00 00 00 00

0060 00 00 00 00 00 00 00 00 00 00 00 00 00 00

WIRESHARK FILTER GENERATOR

Quickly generate Wireshark filters through simple forms.

TCP

HTTP

WEBSOCKET

IP ADDRESSES

PORTS

Source IP

Source Port

Destination IP

Destination Port

Activate Windows

Go to Settings to activate Windows.

Screenshot 3

PCAP VIEWER ONLINE

WASH READY

// LOCAL_PARSING_ONLY

[FORK_ON_GITHUB]

Close File

0 / 120 packets shown

dns

No.	Time	Source	Destination	Protocol	Length	Info
-----	------	--------	-------------	----------	--------	------

▶ Frame 1: 1506 bytes on wire (12048 bits), 1506 bytes captured (12048 bits) on interface en0, id 0 (outbound)

▶ Ethernet II, Src: Apple_1e:38:d7 (64:2b:8c:1e:38:d7), Dst: NECPlatforms_70:c4:08

0020 1a ee 0a 3c 0a 3c 05 c0 05 10 05 00 11 11 00

0030 fe fe fe fd fd fd fd 12 34 56 78 00 00 00 00

0040 00 00 00 00 00 00 00 00 00 00 00 00 00 00

0050 00 00 00 00 00 00 00 00 00 00 00 00 00 00

0060 00 00 00 00 00 00 00 00 00 00 00 00 00 00

WIRESHARK FILTER GENERATOR

Quickly generate Wireshark filters through simple forms.

TCP

HTTP

WEBSOCKET

QUERY INFO

RESPONSE INFO

Query Name

Response Code

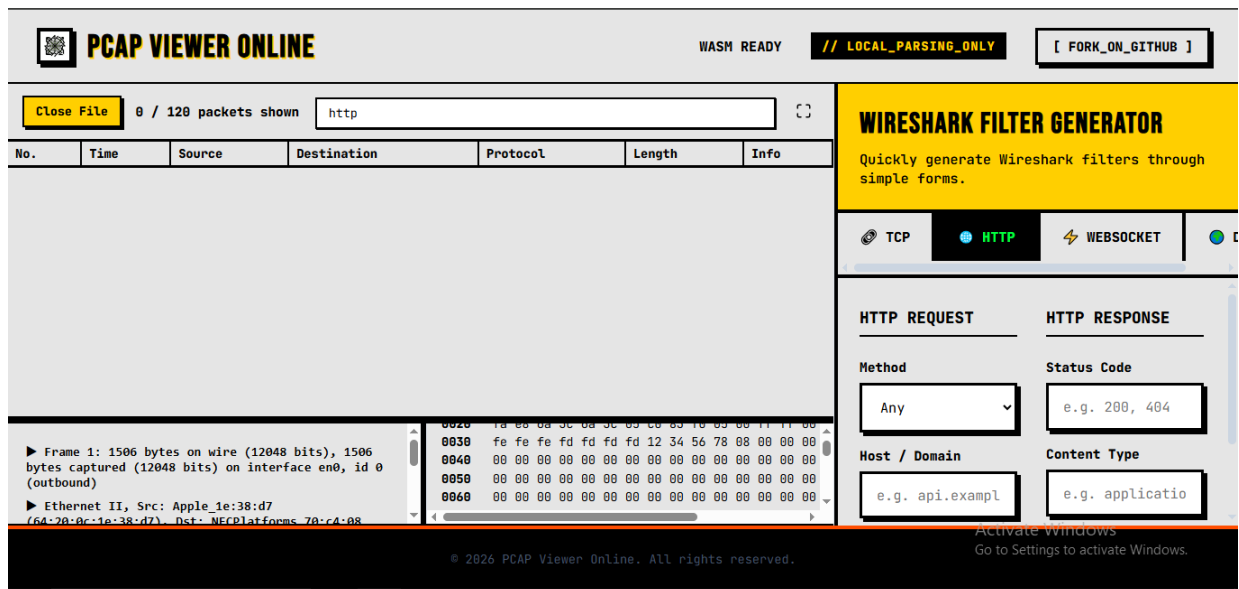
Query Type

Answer Contains

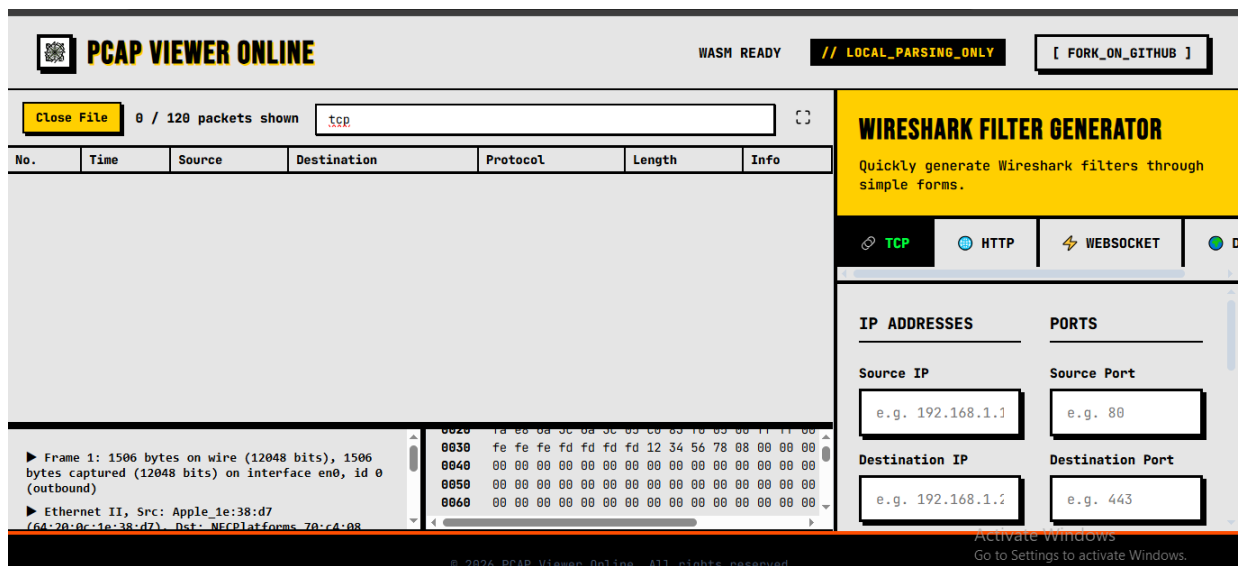
Activate Windows

Go to Settings to activate Windows.

Screenshot 4



Screenshot 5



5. Wireshark Display Filters

Display filters allow an analyst to narrow down a large packet capture and focus on specific traffic.

5.1 IP Address Filter

The following filter was used:

```
ip.addr == 192.168.1.6
```

This filter displays packets where 192.168.1.6 is either the source or destination IP address.

This is useful when investigating communication involving a particular host.

5.2 DNS Filter

The following display filter was used:

`dns`

This filter displays DNS-related packets.

DNS analysis can help an analyst investigate:

- Domain lookups
- DNS request frequency
- Suspicious domains
- Unexpected destinations
- Potential command-and-control activity

5.3 HTTP Filter

The following display filter was used:

`http`

This filter displays HTTP traffic.

HTTP traffic can contain useful information such as:

- HTTP requests
- HTTP responses
- Hostnames
- URLs
- HTTP methods
- User-Agent information
- Transmitted data

Because HTTP is normally unencrypted, sensitive information may potentially be visible within the packet contents.

5.4 TCP Filter

The following display filter was used:

`tcp`

This filter displays TCP packets.

TCP analysis can help identify:

- Network connections
- Source and destination ports
- TCP flags
- Connection establishment
- Connection termination
- Communication patterns

6. Network Traffic Concepts

IP Addresses

IP addresses identify hosts participating in network communication.

During a security investigation, filtering traffic by IP address can help determine what a particular system is communicating with.

For example:

`ip.addr == 192.168.1.6`

can be used to isolate traffic involving a specific host.

DNS

The Domain Name System (DNS) converts domain names into IP addresses.

For example:

`example.com → IP address`

DNS traffic is valuable during investigations because suspicious domains, repeated lookups, unusual domains, and unexpected DNS destinations may provide indicators of compromise.

HTTP

HTTP is an application-layer protocol used for web communication.

HTTP traffic is generally transmitted without encryption, meaning information contained in requests and responses may be visible during packet inspection.

This makes HTTP traffic useful when investigating potentially exposed credentials, URLs, parameters, or other transmitted information.

TCP

Transmission Control Protocol (TCP) provides reliable communication between network hosts.

TCP analysis can help an analyst understand:

- Which systems communicated
- Which ports were used
- When connections were established
- How connections were terminated
- What application protocols may be carried over the connection

7. HTTP Request Analysis

An HTTP packet was inspected during the practical exercise.

When analyzing an HTTP request, a SOC analyst can examine fields such as:

- Source IP
- Destination IP
- Source Port
- Destination Port
- HTTP Method
- Host
- Request URI
- User-Agent
- Packet Data

For example, an HTTP request may contain:

GET / HTTP/1.1

Host: example.com

Because HTTP is not encrypted by default, packet inspection can potentially reveal information transmitted between the client and server.

Security Relevance

Plaintext HTTP can expose information that would otherwise be protected when using encrypted protocols such as HTTPS.

Potentially exposed information can include:

- URLs
- HTTP parameters
- Session information
- Application data
- Credentials, if an insecure application transmits them over HTTP

8. Understanding Suspicious Network Traffic

A SOC analyst should not assume that every unusual packet represents malicious activity.

Traffic should be investigated in context.

Potential indicators that may require further investigation include:

- Unexpected external IP addresses
- Repeated connections to an unfamiliar destination
- Unusual destination ports
- Large amounts of unexpected traffic
- Suspicious DNS queries
- Communication with known malicious infrastructure
- Cleartext credentials
- Unusual HTTP requests
- Repeated failed connections

These indicators should be correlated with other security data before determining whether an event is malicious.

9. SOC Analyst Relevance

Network traffic analysis is directly relevant to Security Operations Center work.

Packet analysis can assist during investigations involving:

- Suspicious IP addresses
- Unexpected network connections
- Malware communication
- Command-and-control traffic

- Suspicious DNS requests
- Unencrypted HTTP traffic
- Possible credential exposure
- Unusual ports and protocols
- Data exfiltration investigations
- Incident response

A SOC analyst can combine packet-capture evidence with:

- Firewall logs
- DNS logs
- Endpoint logs
- Authentication logs
- Proxy logs
- SIEM alerts
- Threat intelligence

This provides a broader view of an incident.

10. Practical Skills Demonstrated

11. Evidence Summary

The screenshots included in this report provide evidence of the practical work completed for the project.

Evidence Files

w1.png

w2.png

w3.png

w4.png

w5.png

The evidence covers the required learning and practical network traffic analysis activities.

12. Repository Structure

Week-3-Task-9-Network-Traffic-Analysis-Wireshark-Basics/



■■■ README.md

■■■ Week-3-Task-9-Network-Traffic-Analysis-Wireshark-Basics.pdf

■

■■■ Report/

■■■ report.md

■■■ w1.png

■■■ w2.png

■■■ w3.png

■■■ w4.png

■■■ w5.png

13. Conclusion

This project provided practical exposure to network traffic analysis using Wireshark and packet capture data.

The main skills developed were:

- Understanding packet captures
- Identifying network protocols
- Filtering traffic by IP address
- Filtering DNS traffic
- Filtering HTTP traffic
- Filtering TCP traffic
- Inspecting HTTP requests
- Understanding plaintext information exposure
- Recognizing network indicators relevant to SOC investigations

Network traffic analysis is an important foundation for SOC analysts because packet-level evidence can help explain what happened during a security event and support investigation and incident-response activities.

14. Final Status

Project: Week 3 — Task 9: Network Traffic Analysis — Wireshark Basics

Status: Completed

Practical Area: Network Traffic Analysis

Primary Tool: Wireshark

Learning Activity: TryHackMe — Wireshark: The Basics

Report: Completed with practical evidence and screenshots